Lesson 12B Elasticity

Elasticity

Although we often idealize rigid bodies as undeformable, all real materials are elastic to some extent—that is, their shape or size changes in response to applied forces. These deformations are governed by the concepts of **stress** and **strain**, and their relationship defines a material's elastic behavior.

Stress and Strain

- **Stress** is the external force applied **per unit area**:

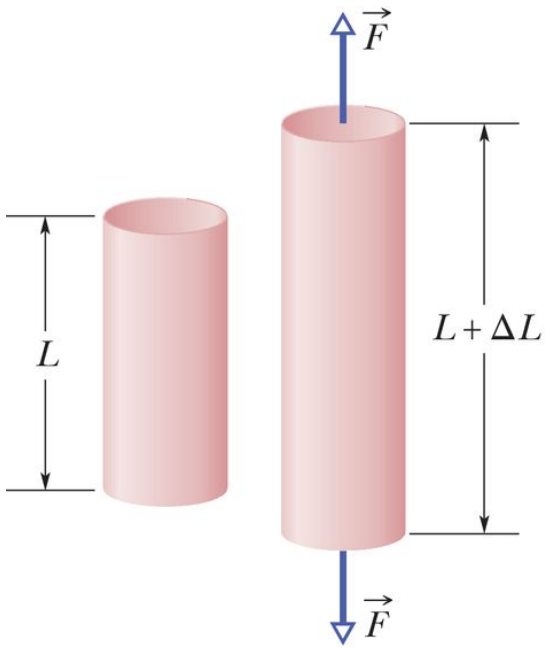
$$\text{Stress} = \frac{F}{A}$$

  where F is the applied force and A is the cross-sectional area over which the force acts.

- **Strain** is the resulting **unit deformation**, which is dimensionless:

$$\text{Strain} = \frac{\Delta L}{L}$$

  where ΔL is the change in length and L is the original length as shown in the following image.



In the **elastic region**, stress and strain are proportional:

$$\text{Stress} = (\text{Elastic Modulus}) \times \text{Strain}$$

This relation is known as **Hooke's Law**, and it varies depending on the type of deformation.

Types of Stress and Their Corresponding Moduli

There are three primary types of mechanical stress, each associated with a specific modulus of elasticity:

1. Tension or Compression (Linear Deformation)

- Stress: $\frac{F}{A}$
- Strain: $\frac{\Delta L}{L}$
- Elastic Modulus: **Young's Modulus** E

$$\frac{F}{A} = E \cdot \frac{\Delta L}{L}$$

Materials have the same modulus E for both tension and compression, but the **strengths** can

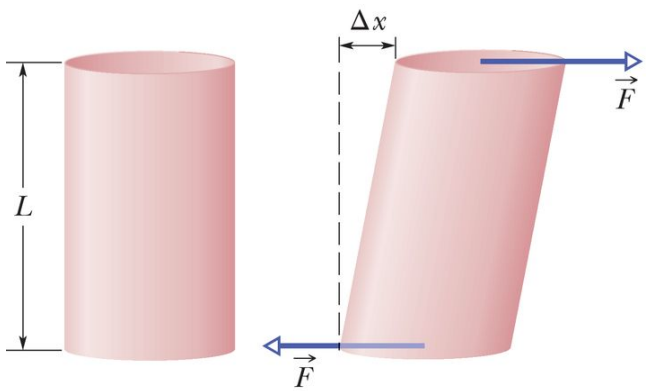
differ widely. For example, concrete has high compressive but low tensile strength.

2. Shearing (Lateral Deformation)

- Stress: $\frac{F}{A}$, but here F is **parallel** to the surface
- Strain: $\frac{\Delta x}{L}$, where Δx is lateral displacement and L is the height
- Elastic Modulus: **Shear Modulus** G

$$\frac{F}{A} = G \cdot \frac{\Delta x}{L}$$

Shearing occurs when forces are applied tangentially to a surface, such as when using scissors or applying torsion to a rod. This is shown in the image below.

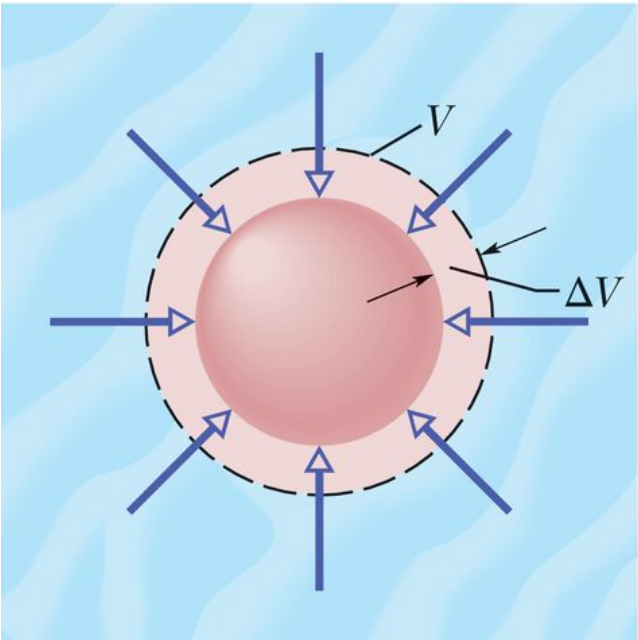


3. Hydraulic Compression (Volume Deformation)

- Stress: Pressure $p = \frac{F}{A}$
- Strain: $\frac{\Delta V}{V}$, where ΔV is the change in volume
- Elastic Modulus: **Bulk Modulus** *B*

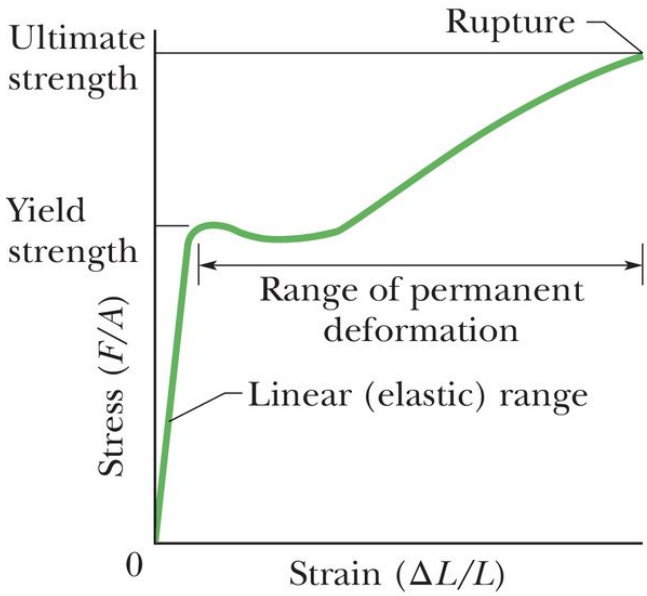
$$p = B\frac{\Delta V}{V}$$

The bulk modulus describes a material's resistance to uniform compression, such as being submerged in water as shown below.



Elastic Limit and Failure

As stress increases, materials eventually reach a **yield point**, beyond which deformation becomes **permanent**. If stress continues to increase, the material reaches its **ultimate strength**, at which point it fractures. The tensile test is a standard test of a materials tensile properties and produces a stress-strain curve like that shown in the figure below.



The **stress-strain curve** illustrates this progression:

- **Linear region**: elastic behavior, obeys Hooke's law
- **Yield point**: start of permanent deformation

- **Ultimate strength**: maximum stress the material can handle
- **Fracture point**: material breaks

Tables of **elastic moduli** for common materials (metals, polymers, biological tissues, etc.) help engineers and physicists choose materials for specific applications based on how they respond to different types of stress.

Table 12-1 Some Elastic Properties

	Density ρ (kg/m^3)	Young's modulus E (10^9 N/m^2)	Ultimate strength S_u (10^6 N/m^2)	Yield strength S_y (10^6 N/m^2)
	ρ	E	S_u	S_y
Steel[a]	7860	200	400	250
Aluminum	2710	70	110	95
Glass	2190	65	50[b]	—
Concrete[c]	2320	30	40[b]	—
Wood[d]	525	13	50[b]	—
Bone	1900	9[b]	170[b]	—
Polystyrene	1050	3	48	—

[a]Structural steel (ASTM-36).
[b]In compression.
[c]High strength.
[d]Douglas fir.